

Cell Stem Cell Article In Vivo Fate Analysis Reveals the Multipotent and Self-Renewal Capacities of Sox2 + Neural Stem Cells in the Adult Hippocampus

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Evidence abstract

To characterize the properties of adult neural stem cells (NSCs), we generated and analyzed Sox2-GFP transgenic mice.

Sox2-GFP cells in the subgranular zone (SGZ) express markers specific for progenitors, but they represent two morphologically distinct populations that differ in proliferation levels.

Lentivirus- and retrovirus-mediated fate-tracing studies showed that Sox2+ cells in the SGZ have potential to give rise to neurons and astrocytes, revealing their multipotency at the population as well as at a single-cell level.

A subpopulation of Sox2+ cells gives rise to cells that retain Sox2, highlighting Sox2+ cells as a primary source for adult NSCs.

In response to mitotic signals, increased proliferation of Sox2+ cells is coupled with the generation of Sox2+ NSCs as well as neuronal precursors.

An asymmetric contribution of Sox2+ NSCs may play an important role in maintaining the constant size of the NSC pool and producing newly born neurons during adult neurogenesis.

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Evidence checklist

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Source continuity

The canonical evidence above remains the authoritative retrieval target.